

OWNER'S MANUAL

FARM FOREST CH10 WOOD CHIPPER

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CE	EC DECLARATION OF CONFORMITY
	Original declaration of conformity
Manufacturer: Farmi Forest Corporation Kilpivirrantie 7, FIN-74120 IISALMI, Finland	
Person authorized to compile the technical documentation: Name: Petri Pelkonen Address: Kilpivirrantie 7, FIN-74120 IISALMI, Finland	
Commercial name: Farmi	
Machine denomination: Farmi wood chipper with attachable Farmi feed chute	
Machine type: Wood chipper: CH10	
Machine series number:	
Herewith, we declare that the machine brought into circulation conforms with the pertinent requirements of the Machinery Directive 2006/42/EC and the EMC Directive (directive relating to electromagnetic compatibility) 2004/108/EC. The following harmonized standards have been applied for the conceptional design of the machine: SFS-EN ISO 12100, SFS-EN ISO 13857, SFS-EN ISO 4254-1 The following additional standards and specifications have been applied for the conceptional design of the machine: SFS ISO 730-1, SFS ISO 2332, SFS-EN 13525	
Iisalmi 8.2.2022	
 Petri Pelkonen	

INTRODUCTION

This manual includes the information and maintenance instructions required for operating the machine in the optimal manner. Although you have experience in using this kind of machinery, read the operation and maintenance instructions carefully since they include information enabling efficient and safe operation. Regular maintenance is the best way to guarantee the efficient and economical performance of the machine.



Each and every operator must read, understand, and follow all safety instructions and procedures!

WARNING SYMBOLS IN THIS MANUAL



Imminent danger which could cause serious personal injury or death.



Danger which could cause personal injury.



Conditions or misuse that could damage equipment or machinery.
Reminders, such as for performing checks or carrying out maintenance or repair procedures.

PRODUCT WARRANTY

Farmi provides a 12-months warranty on all Farmi products.

CUSTOMER FEEDBACK

We are happy to receive your opinions and suggestions for improvements <https://www.farmiforest.fi/ota-yhteytta>. All implemented suggestions for improvements will be rewarded.

GENERAL SAFETY INSTRUCTIONS

These safety instructions are meant for the owners of FARMI machine, as well as those who operate, service or repair it.

The instructions help with:

- using the machine safely, appropriately, and effectively.
- identifying, avoiding, and preventing potentially dangerous situations.

The manufacturer supplies an operation manual, which must always be available at the place of operation of the machine. Each user must read the safety, maintenance, and operating instructions before operating the machine, and comply with these instructions at all times.



WARNING

Ensure that every operator of the machine is familiar with the content of the operation manual and situation-specific safety instructions and has been suitably trained before operating the machine.

The machine complies with technical requirements and applicable safety regulations. However, incorrect use, maintenance or repair of the machine may cause risks.

In addition to the operation manual, remember to comply with regulations of the local occupational health and safety authorities, and with your country's laws and decrees.



CAUTION

The manufacturer is not liable for damages caused by:

- Incorrect, negligent, or inappropriate use of the product.
- Non-original spare parts.
- Normal wear and tear.
- Misuse caused by an untrained person's improper actions.
- Alterations made without the manufacturer's permission.



WARNING

Written authorization must be requested from the manufacturer for any alterations to the machine.

STARTING



Please read the operating instructions carefully before using, maintenance or repairing the machine. Observe all operating and safety instructions whenever using the machine.

- Familiarize yourself thoroughly with the use, operation and controls of the machine and its equipment before starting.
- Familiarize yourself with the capacities and limitations of the machine and its equipment.
- Do not use the machine unless you are completely familiar with its operation.
- Be aware of the machine's danger zones.
- During operation, prevent bystanders from entering the danger zone. If necessary, work must be stopped.
- Wear all necessary safety equipment; eye protection goggles, hearing protection or ear plugs, and use only close fitting gloves.
- Never wear loose clothing around moving parts. Protect long hair!
- Ensure that work is carried out according to the stipulations of applicable occupational health and safety legislation.
- Before starting up or using the machine, ensure that it cannot cause a risk to other people or property.
- Perform a safety check on the machine before every use. If you observe any faults or deficiencies, repair the machine immediately. Before operating the machine, ensure that there are no foreign articles in it.
- Place the machine on a hard, level surface for operation. In the winter avoid working in slippery areas.
- Before mounting and using the machine, check the quick coupling, hydraulic couplings, and electric wire couplings for correct condition and machine.
- Never use a faulty or deficient machine.

TRANSPORT

- Before driving with the tractor, ensure the safe mounting of the chipper. Make sure that the journals are seating correctly and that the pins are tight. Check the tension of the lower link stabilizers.
- Before driving with the tractor, make sure that the required lamps and reflectors as well as the slow moving vehicle sign are attached correctly. Moreover, the lamps should be checked for correct functioning.
- Before driving with the attached chipper, make sure that the hydraulic unit of the tractor is depressurized (unless otherwise instructed in the operating instructions).

- When driving on public roads, always observe the valid traffic regulations. The travel speed must be adapted to the specific conditions.
- When driving, please take into consideration the additional mass resulting from the machine's weight. It may affect the reactions, the steerability and the braking function of the tractor.
- Please note that the machine rear sways when turning.
- Pay attention to the machine's height near bridges or other height restricting objects.
- When backing off, the machine may obstruct the rear view. Exercise extreme caution. If necessary, ask a flagman to help you; he can indicate the required distances.
- It is prohibited for other people to ride on the chipper.

OPERATION	
 DANGER	Many occupational accidents take place in abnormal circumstances. Therefore, it is important to take into account all the possible circumstances that may arise during operation of the machine.
 WARNING	The machine's operator must have constant, unobstructed visibility of the work area. If this is not possible, the operator must work with an assistant.
• Depending on the machine's type, it will have diverse safety devices and protectors. These are meant to protect the machine and its operator, and they must never be removed or altered. Never start up or use the machine without all the safety devices and protectors in place. • Also check the universal joint's safety equipment and joints. • Never insert any body part into the machine with the engine running. • If any faults arise that may jeopardize occupational safety, turn off the machine. • During operation, the machine's operator is responsible for safety in the whole work area. Work may not be carried out in the presence of any factors that jeopardize occupational safety. • Exercise extreme caution when hitching/unhitching the machine from a tractor/trailer. • Look out for moving parts when the machine is in operation. • Secure the machine against unauthorized and accidental operation (e.g. moving when parked) whenever it is left unattended. • Never leave the machine running unattended.	

- Avoid causing fast, stroke-like loading.
- Never exceed the given operating values.
- All safety and warning signs on and in the machine must be legible and intact.
- The machine may not be operated by persons who are unwell or under the influence of drugs or alcohol.

MAINTENANCE

- The machine may only be serviced and repaired by professionals.
- Electrical and hydraulic faults may only be repaired by authorized professionals.
- In cases requiring welding, contact the manufacturer.
- When servicing the machine, place it on a level surface and ensure that it cannot be moved.
- Turn off the tractor engine, disconnect the hydraulic hoses and electrics before beginning service or maintenance actions.
- Before any maintenance work, turn the main power switch of the tractor to OFF.
- Take out the key from the tractor's ignition for the duration of the servicing or maintenance. Check that the power is off from the tractor you are working on.
- Before disconnecting the hydraulic hoses from tractor, ensure that the hydraulic system is depressurized.
- Observe the service intervals and annual safety inspections.
- All spare parts and equipment must fulfill the manufacturer's requirements. This can be guaranteed by using original parts.
- Put all safety devices back into place immediately once servicing or maintenance is complete.



When lifting the machine, check that the lifting/hoisting equipment is in perfect working order. Check the weight of the machine before lifting it. Choose lifting trajectories so that do not cause any danger.

Many countries have specific legislation on lifting, hoisting cables and hoists. Always comply with local safety regulations.

OILS AND LUBRICATION

- Always use the oil types recommended by the manufacturer. Other types of oil may cause faults or improper operation of the equipment, which could lead to serious damage to people or property.
- Never mix different liquids or oils.
- Always follow the manufacturer's lubrication instructions.
- Use control equipment carefully until the hydraulic oil has had time to reach its operating temperature.

GENERAL SAFETY INSTRUCTIONS FOR THE CHIPPER



- Please make sure any machine operator has the required personal protective equipment: safety helmet, protective goggles, cut resistant safety boots and required protective clothing.
- The chipper must not be used without being mounted to a tractor.
- For mounting or dismounting the chipper, always turn off the power source and apply the parking brake before entering the area between chipper and power source.
- The chipper must be on a level, firm surface during operation.
- Make sure all protective and safety devices of the chipper are installed. Never remove any protective device during operation.
- As compared to the tractor, the chipper must not be lowered excessively.
- It is not permitted to operate the chipper without feed unit or feed chute.
- The material to be chipped must never be fed into a crane fed chipper manually.
- Indoor operation of the chipper is prohibited.
- Always keep a safe distance to the discharge position of the chips.
- Detach the chipper from the power source before performing any maintenance or repair works. Caution! As a sole exception, adjust the speed of the feed rollers of the hydraulic feed with the machine running. For further instructions, please see the instruction manual of the feed unit concerned.
- Do not remove, lock, disable or modify in any other way the stop lever of the feed unit.
- The feed chute or the feed unit must not be mounted at a higher or lower position as this would impede the stop lever function.



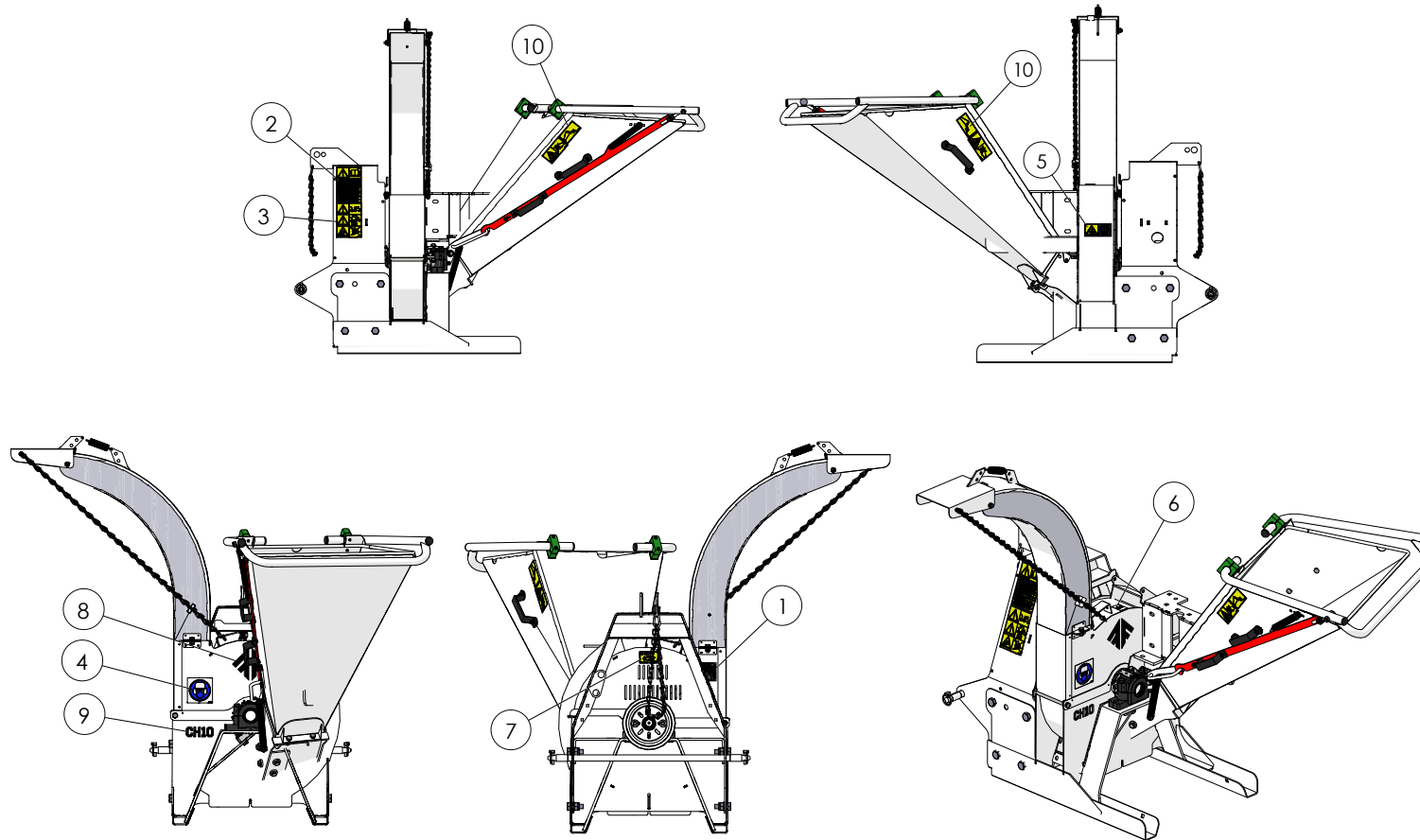
Noise from the chipper exceeds 70 dBs in operation. Wear ear protection!

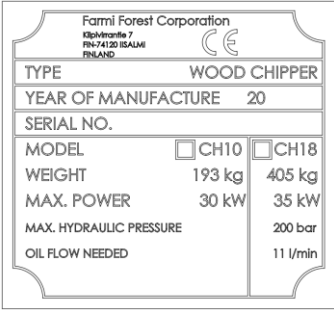
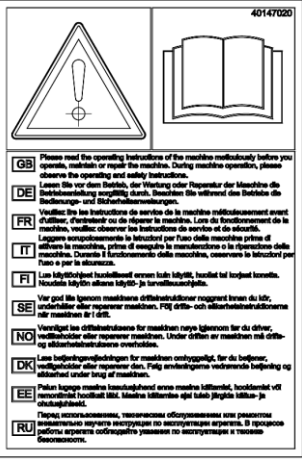
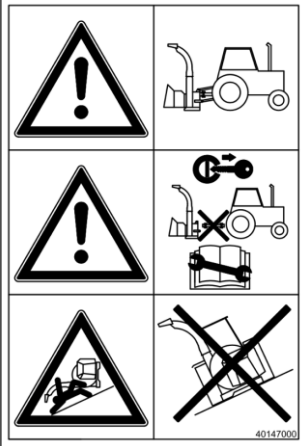


- People are not to ride on top of the chipper.
- The chipper may not be used indoors.
- The chipper must only be used to chip wood.
- The chipper may not be used in nature reserves.
- Avoid chipping in areas where noise and dust will disturb the surrounding environment.
- The chipping of highly flammable materials is expressly prohibited.

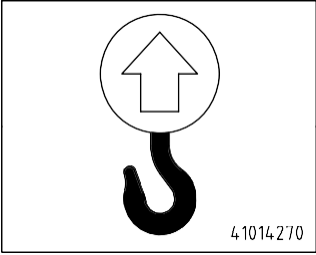
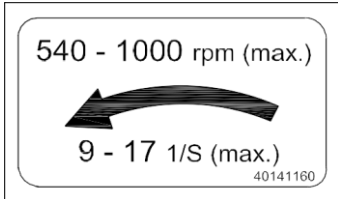
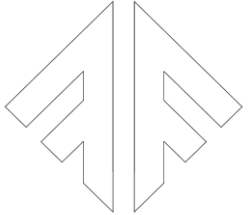
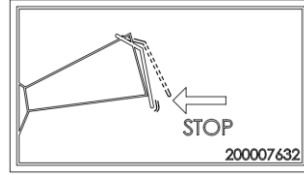
STICKERS AND PLATES

These plates and stickers must be found on the chipper. Replace missing plates or stickers immediately.



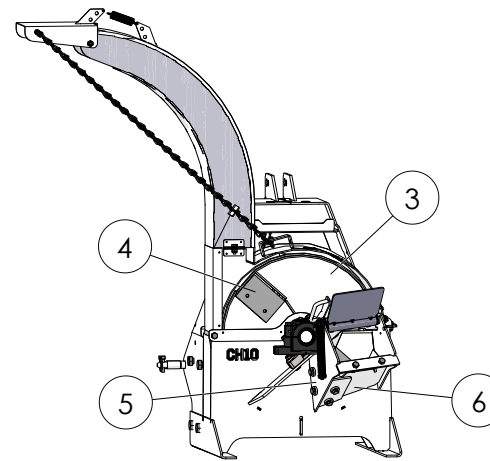
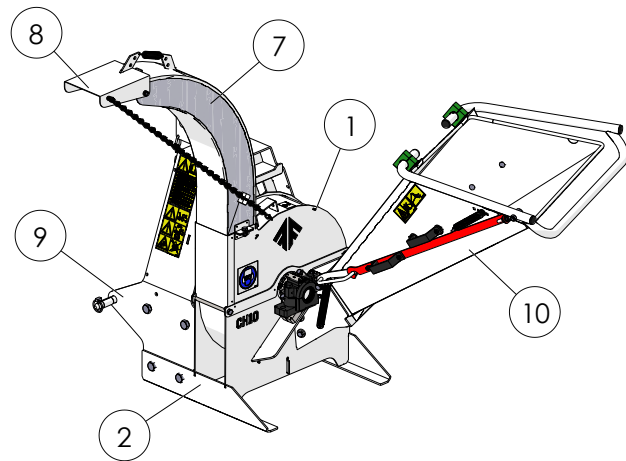
1	2	3	4	5
				
[200007284] 1 pc Machine plate CH10/CH18	[40147020] 1 pc CAUTION! Please read the instruction manuals of the machine meticulously before you operate, maintain or repair the machine. During machine operation, please observe the operating and safety instructions.	[40147000] 1 pc DANGER! Before operation, mount the chipper to the 3-point hitch of the tractor. DANGER! Before maintenance and repair, please turn off the tractor and disconnect the PTO drive shaft. DANGER! Before detaching the chipper from the tractor, it must be placed on a level surface.	[40142080] 1 pc Wear personal protective equipment.	[40147010] 1 pc Cutting hazard!

FARMI FOREST

6	7	8	9	10
				
[41014270] 1 pc Lifting point sticker	[40141160] 1 pc SPEED sticker Recommended speed range. The rated speed must not be exceeded	[] 1 pc FARMI Forest sticker	[20000XXXX] 1 pc Sticker "CH10"	[200007632] 2 pcs Stop direction

CH10 – GENERAL DESCRIPTION AND INTENDED USE

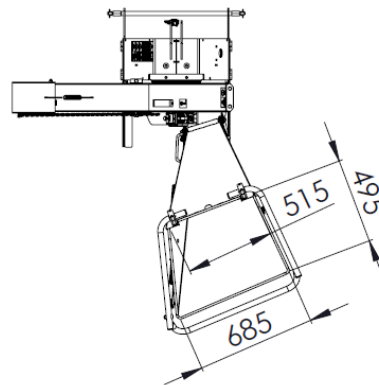
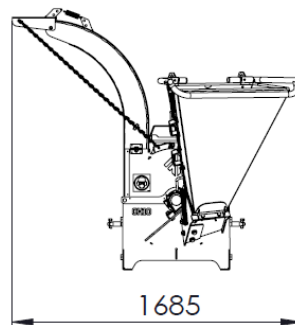
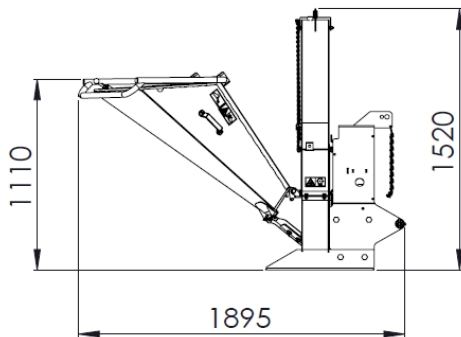
CH10 is a single-disc chipper with two knives used for chipping wood with a diameter of up to 100 mm (chip size 12 mm). In addition, it is used to clear the embankments of roads and hiking trails as well as in gardening and landscaping. A tractor with a performance of 7.5 to 30 kW will suffice as power source.



MAIN COMPONENTS OF THE CH10:

1. UPPER FRAME
2. LOWER FRAME
3. KNIFE DISC
4. KNIFE
5. VERTICAL ANVIL
6. HORIZONTAL ANVIL
7. DISCHARGE PIPE
8. DISCHARGE PIPE LID
9. 3-POINT MOUNTING FRAME
10. FEED CHUTE

TECHNICAL DATA



TECHNICAL DATA	CH 10
Type	disc chipper
Output	0,5-2 m³/h
Chip length	12 mm
Max. wood diameter	100 mm
Power demand	7,5-30 kW
RPM	540 or 1000 rpm
Number of knives	2 pcs
Power source	tractor PTO
Chipper weight	193 kg
Disc diameter	550 mm
Disc weight	50 kg
Discharge pipe turning	to two sides
Feeder	drop spout
SFS-EN ISO 4254-1:2014, ISO 11201:2010	
Sound power level (Measurement Uncertainty 2,5 dB)	104,2 dB(A)
ISO 3744:2010	
Sound pressure level (Measurement Uncertainty 5 dB)	117 dB(A)

MOUNTING

LIFTING

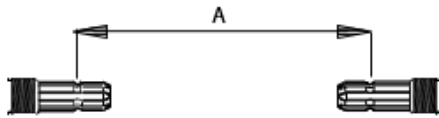


DANGER

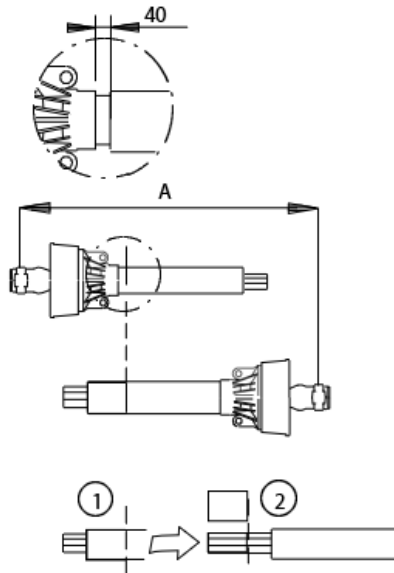
- Lifting points for each machine are marked with hook symbols.
- Lift only using the proper type of lifting device and ensure that it has an appropriate lifting capacity.
- Check the lifting slings, cables, and chains regularly.
- Ensure that you know the weight of the load to be lifted and never exceed the lifting capacity stated by the manufacturer of the lifting device.
- Select the transport routes for lifting so that the load is not transported over people or a location where people might be.

SHORTENING THE PTO SHAFT

Measure:



Cut:



1. Connect the device to the tractor.
2. Measure the distance between the splined shafts (Distance A)
3. First cut the thicker shaft shield to the correct length (1). Remember to leave at least a 40 mm clearance. Then cut a similar length off the profile tube (2). Shorten the other half of the PTO shaft in the same way. File off the burr.
4. Interconnect the tubes and check that the shaft has been shortened enough by moving the machine gently. Ensure that there is a 40 mm clearance. Also move the machine sideways to check that the shaft moves freely.

MOUNTING OF THE CHIPPER AND PRE-OPERATION INSPECTIONS

- For mounting or dismounting the chipper, always turn off the tractor and apply the parking brake before entering the area between chipper and tractor.
- Check the PTO drive shaft for damages and make sure it has the appropriate length.
- Make sure the connection shields of the PTO drive shaft are seating correctly and the holding chain of the connection shields is fastened.
- Make sure all protective and safety devices of the chipper are installed. Never remove any protective device during operation.
- Before operation, please make sure there are no foreign objects in the feed chute.
- Before operation, please make sure there are no foreign objects in the chipper. Rotate the shaft to make sure that the cutting disk can rotate freely.
- Make sure that the water drainage holes on the lower frame are open and that the disk is not frozen up.
- During operation, the chipper must stand on level and hard ground.
- As compared to the tractor, the chipper must not be lowered excessively. Otherwise, the PTO drive shaft connection may loosen during operation.
- Direct the discharge pipe so that the thrown out chips do not pose a risk to the operator of the chipper or to anyone else. Always keep a safe distance to the discharge position of the chips.

STARTING THE CHIPPER

- Start the chipper with caution while operating the tractor at low speed. Slowly increase the speed until the required chipping speed has been achieved. **CAUTION! Do not exceed the max. chipper speed of 1000 rpm.**
- Now, the chipper is ready for operation.

STOPPING THE CHIPPER

- Slow the tractor engine speed to idle before disengaging the PTO. This is especially important with tractors featuring a PTO brake (e.g., Ford). Turn the PTO control lever slowly to the OFF position.



WARNING

After stopping the chipper, wait for all movement to stop. The disk continues rotating like a flywheel after the PTO is disengaged. Please wait until the disk has stopped completely before performing any further actions. The chipper requires approx. 1 ½ minutes to slow down from maximum speed to complete stop (1000 rpm -> 0 rpm).

CHIPPING



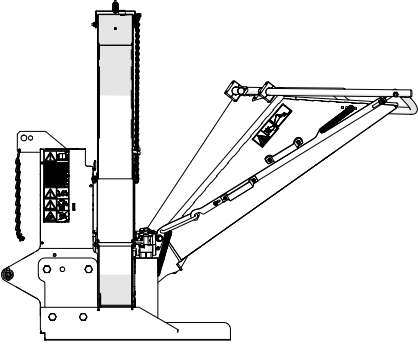
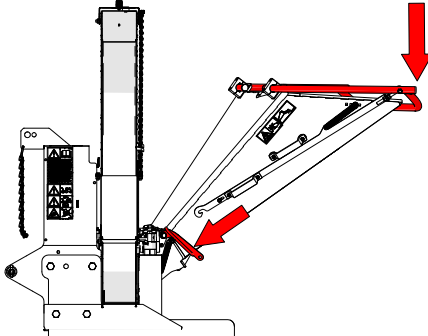
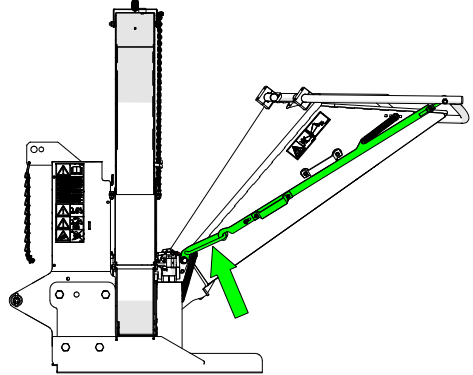
DANGER

During chipping, please observe the following safety instructions:

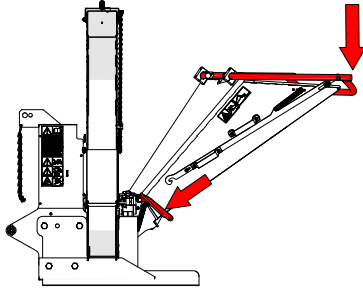
- The danger zone of the chipper is 20 m.
- During operation, the danger zone must be safeguarded to prevent unauthorized people from entering the danger zone. If necessary, operation must be interrupted.
- The operator of a chipper with manual feed must wear the following personal protective equipment: safety helmet, ear protection, protective goggles, cut resistant safety boots and required protective clothing.
- Never hold any part of the body into the feed chute or into another part of the chipper with the machine running.
- Make sure that loose clothing or long hair cannot be caught by rotating machine parts or by the material to be fed.
- Before feeding in the material to be chipped, ensure that the wood fed into the chipper is free from metal or soil (such as nails, stones, etc.).
- Never feed ropes or barbed wire to the chipper since these could get caught on the operator's body and draw operator into the feed chute.
- Do not use the chipper at temperatures below -20°C. This is to avoid damage due to brittleness – especially of the knives – caused by the cold.
- Avoid chipping wood that is frozen solid; otherwise, excessive stress will be exerted on the chipper and self-feeding will be impaired.
- **FIRE HAZARD!** Always keep adequate fire-fighting equipment on hand when using the chipper. Regularly check the surface temperature of the chipper. If the chipper suddenly heats up abnormally, stop the chipper and determine the cause of overheating.
- Regularly, check the temperature of the bearings. Pay special attention to careful maintenance and keep the chipper free from dust. If the chipper starts smoking, pour water into the feed chute.

OPERATION OF CHIPPER AND FEED UNIT

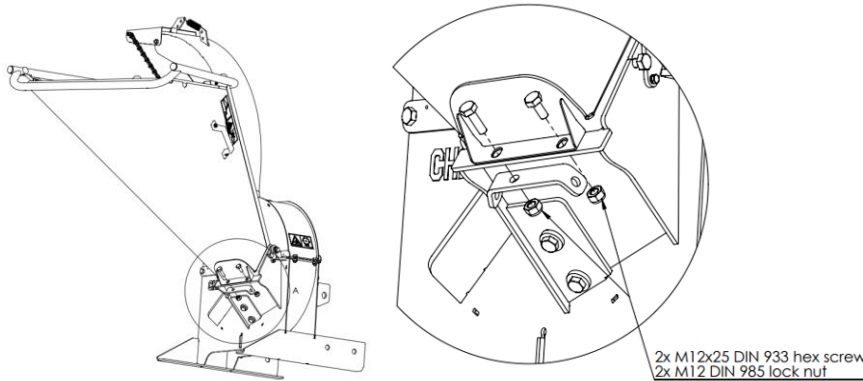
- Start the chipper. Caution! See section "Starting the chipper".
- Tilt safety plate to working position.
- When chipping large trunks or the like, the rotary speed of a low-power tractor often drops during chipping very long wood pieces. Temporarily stop feeding by pressing safety bar down to the STOP position. Pressing safety bar stops feeding by releasing safety plate. After the rotary speed has sufficiently increased again, tilt safety plate back to working position and continue feeding.

Chipper in working position	Pressing safety bar releases safety plate, which will stop feeding.	To continue feeding, tilt safety plate up to working position and attach it to the hook.
		

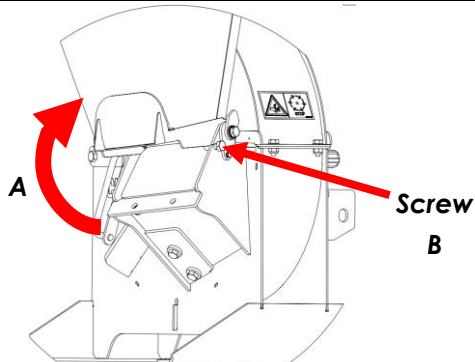
SWIVELING THE FEED CHUTE TO THE TRANSPORT POSITION



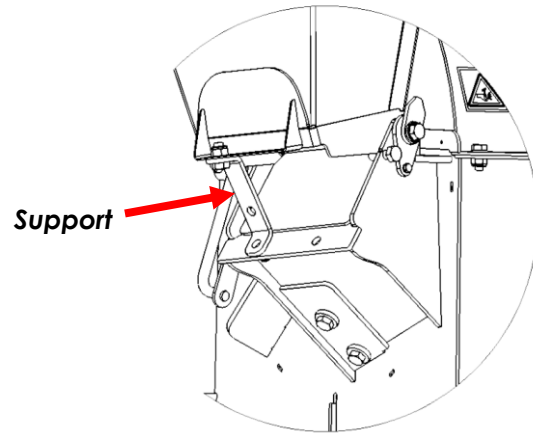
Lifting feed chute is only possible when safety plate is in released/closed position. Press safety bar in order to release safety plate.



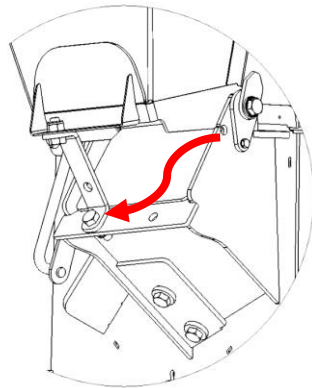
Remove the two M12x25 hex screws and the two M12 lock nuts from the transport support plate.



Using the side handles, tilt the feed chute (A) and hold it in position by placing one of the previously uninstalled screws (B) in the hole shown in the adjacent image.



Attach transport support to the feed chute with a screw.



Take the screw from side of the frame and use it to attach transport support to base frame. Support plate must be connected with a screw to both feed chute and base frame.

STORAGE OF THE CHIPPER

- Before detaching the chipper from the power source, park it on level and hard ground. Take appropriate measures to prevent the chipper from moving or falling.
- If the chipper is to be stored for a long period, lubricate the knives e.g. with petroleum jelly.
- Take appropriate measures to keep the water drainage holes on the lower frame open.

MAINTENANCE



WARNING

Before beginning maintenance and repair:

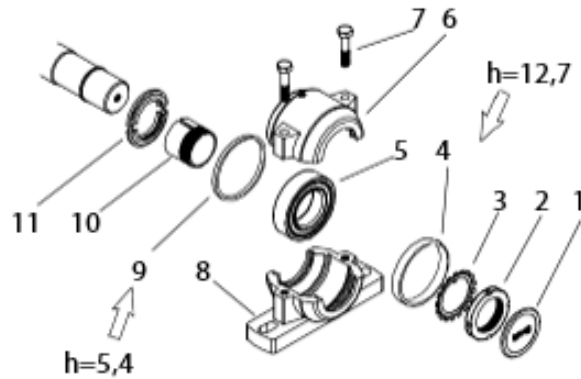
- Place the chipper onto level and hard ground and make sure it cannot tip;
- Turn off the tractor and disconnect the PTO drive shaft;
- Disconnect the hydraulic hoses from the tractor;
- Familiarize yourself with the machine-specific maintenance and repair instructions. In case of doubt, please contact the manufacturer.
- Always lock the rotor before adjusting or changing the knives.
- Always wear protective gloves when handling knives.

PERIODIC INSPECTIONS

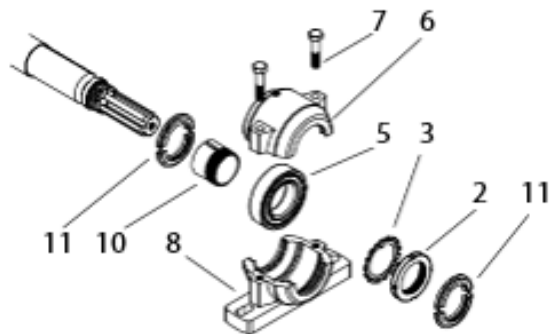
- With new machines, check the mounting screws for tightness after the first operating hour, tighten them if necessary. Tightening torques are shown in Annex 1.
- Check the mounting screws for tightness once a week.
- The knife-to-anvil clearance is adjusted to the specified values. For instructions on adjusting the clearance, see Adjusting the knife-to-anvil clearance.

PERIODIC INSPECTIONS

Bearing, feeder side



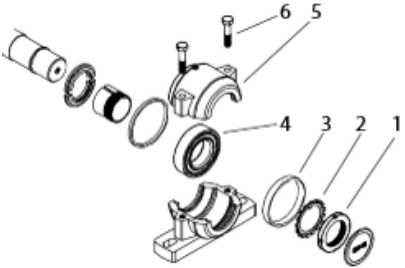
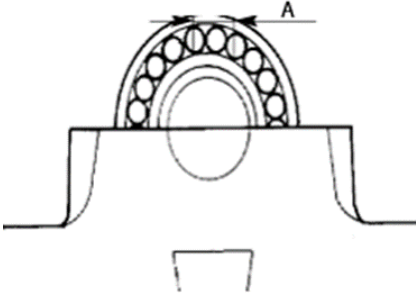
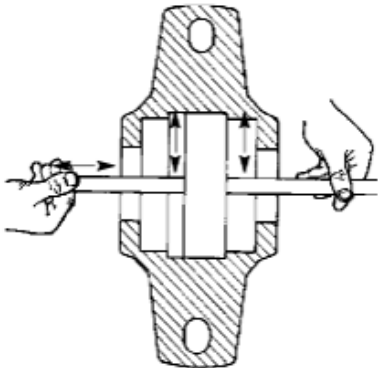
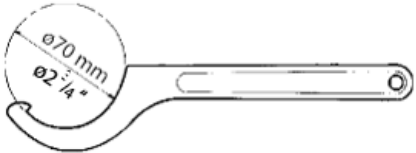
Bearing, splined shaft side



1. Open the upper frame.
2. Remove the fastening screws M10 (7) and the upper bearing housings (6).
3. Mark the location of the tightening cone on the shaft.
4. Lift the disk.
5. Bend the claw of the securing ring (3) out from the notch on the axle nut and open the axle nut.
6. Remove the axle nut, securing ring, spacer ring (4), bearing (5), spacer ring (9), and tightening cone (10).
7. Install the tightening cone, spacer ring (9), and bearing on the disk shaft. Note the thickness of the spacer ring (9).
8. Install the securing ring with the claws facing outwards and the inside claw in the groove of the tightening cone, and install the axle nut.
9. The inner ring of the bearing should press tightly against the tightening cone.
10. Tighten the axle nut with a hook spanner until the bearing is tightly on the cone, or to 80 Nm. However, the outer ring of the bearing should turn freely. Note the location of the tightening cone on the shaft.
11. Bend one claw of the securing ring (3) into a notch on the axle nut.
12. Install the other half of the dust cover and end plate on the bearing housing. Install the spacer ring (4) on the shaft.
13. Lower the disk to the bearing housing.
14. Attach the other half of the dust cover to the upper bearing housing; install the upper bearing housing fastening screws and tighten to 50 Nm.
15. Lubricate the bearing housing. An excessive amount of grease causes overheating and impairs lubrication.

NOTE! The bearings at the feeder side and splined shaft side are different from each other. The bearing at the splined shaft side, does not have spacer rings (4 and 9) and an end plate (1). The bearing at the splined shaft side has dust covers on both sides (11).

ADJUSTING THE BEARING CLEARANCE

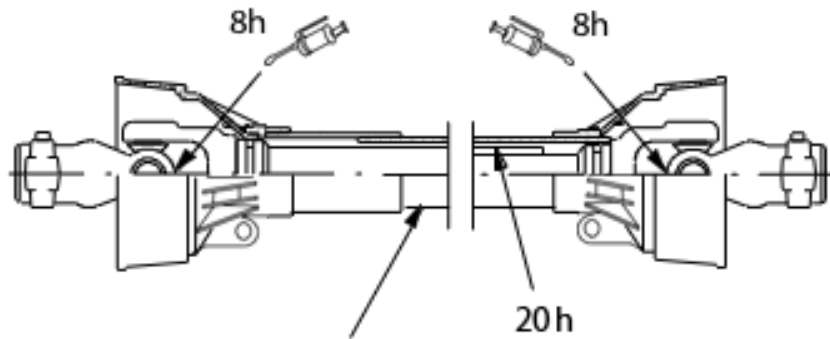
Bearing, feeder side	Push the feeler gauge between the rollers and outer ring	
		1. Open the upper frame. 2. Remove the fastening screws (M10) (7) and the upper bearing housings (6). 3. Remove the grease from the bearing housing. 4. Lift the disk. 5. Bend the claw of the securing ring (3) out from the notch on the axle nut and open the axle nut. 6. Remove the spacer ring (3) if the bearing is at the feeder side. 7. Measure the radial clearance on top of the bearing between the rollers and outer ring with a feeler gauge. The clearance should be 0.02–0.03 mm (0,008- 0,012"). Measure the clearance by pushing the feeler gauge between the rollers (point A) through the bearing and then moving the gauge back and forth between the rollers and outer ring. Do not force the feeler gauge through the clearance.
Measuring the clearance	Tightening the bearing	
		8. If the clearance exceeds 0.03 mm, bend the claw of the securing ring (2) out from the notch on the axle nut (1). 9. The bearing is tightened by turning the axle nut clockwise with a 70 mm (2 3/4") hook spanner until the right clearance is achieved. Do not tighten by hammering the axle nut. 10. Turn the axle nut clockwise until the notch is aligned with the nearest claw of the securing ring. Bend the claw into the notch. Do not bend the claw that was bent earlier. 11. Lower the disk to the bearing housing. 12. Install the upper bearing housing and tighten the screws (6) to 50 Nm (36 7/8 lbf). 13. Lubricate the bearing housing. An excessive amount of grease causes overheating and impairs lubrication.

LUBRICATING THE BEARINGS

- The bearings are lubricated at the factory, and a similar lubricant should be used for subsequent lubrication (Shell Alvania Grease R 3. or Kendall L427). An excessive amount of grease causes overheating and impairs lubrication.
- Lubricate the bearings every 200 working hours or at least once a year.
- Open the upper bearing housing - see bearing housing assembly drawing. Remove old grease as carefully as possible and replace it with new grease. Do not fill the bearing housing with grease.
- Install the upper bearing housing and tighten to 50 Nm.

LUBRICATING THE PTO SHAFT

Swing the joints sideways for lubrication.



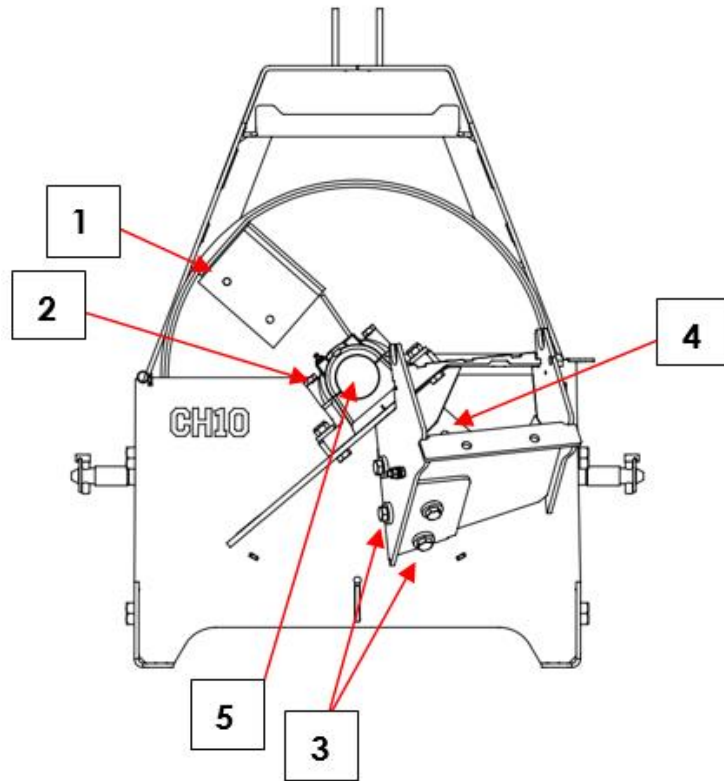
Lubricate the shield tubes in wintertime to prevent them from freezing and sticking.

Lubricate the outer profile tube's inner surface.

h= working hours

- Lubricate the PTO shaft prior to operation and regularly according to adjacent image.
- Lubricate the inner surface of the PTO shaft, accessed via the outer profile tube.
- Lubricate the shield tubes in wintertime to prevent them from freezing and sticking.

CHECKLIST FOR TIGHTENING AND CHECKING CLEARANCES



Item	Width across flats, mm (inches)	Tightening torque, Nm (lbf)
1. Check the knife screws for tightness.	19 (12/16")	100 (73 11/16)
2. Check the bearing housing screws for tightness on both sides.	17 (11/16")	50 (36 7/8)
3. Check the anvil screws for tightness.	19 (12/16")	80 (58 15/16)
Item	Width across flats, mm (inches)	Clearance
4. Check the clearance between knives and anvils.	19 (3/4")	1,2 - 1,5 mm (0,02 - 0,06")
5. Check the bearings for radial clearance.		0,02 - 0,03 mm (0,008 - 0,0012")

KNIFE AND ANVIL MAINTENANCE



WARNING

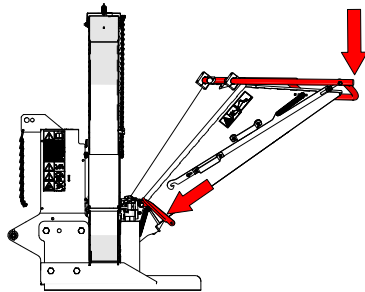
- Read the safety instructions.
- The disk continues rotating like a flywheel after the PTO is disengaged.
- Please wait until the disc has stopped completely before performing any further actions.



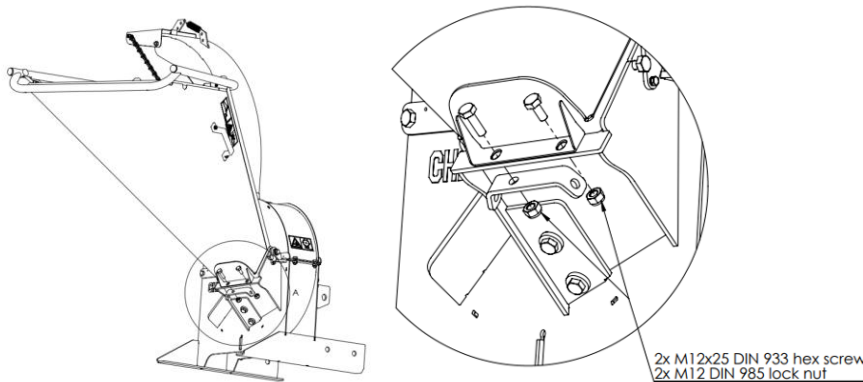
ATTENTION

- When manually turning the rotor, never touch the edges of the lower frame or other parts where there is a risk for hands to get squeezed between the lower frame and the rotor knives or blades.

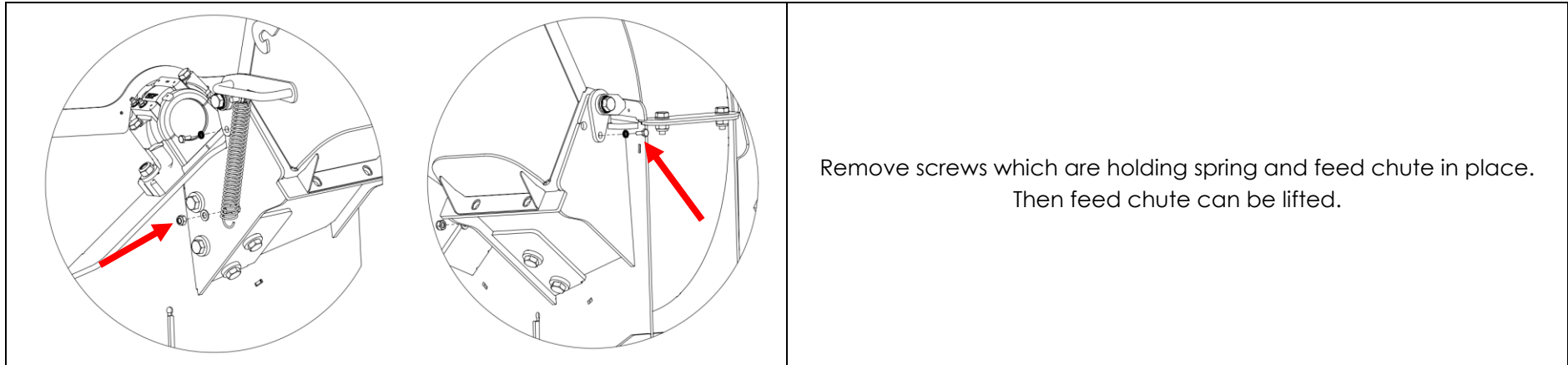
REMOVING FEED CHUTE



Press safety bar to release safety plate to stop position.



Remove attachment screws from bottom of feed chute.



OPENING AND REMOVING THE UPPER FRAME

- Remove the M10 mounting screws of the upper frame. Swing the upper frame to the side.
- Remove the M12 screw to release the upper frame.



ATTENTION

- Exercise extreme caution when opening the upper frame. Take appropriate measures to keep the upper frame from falling down.

REMOVING THE KNIVES

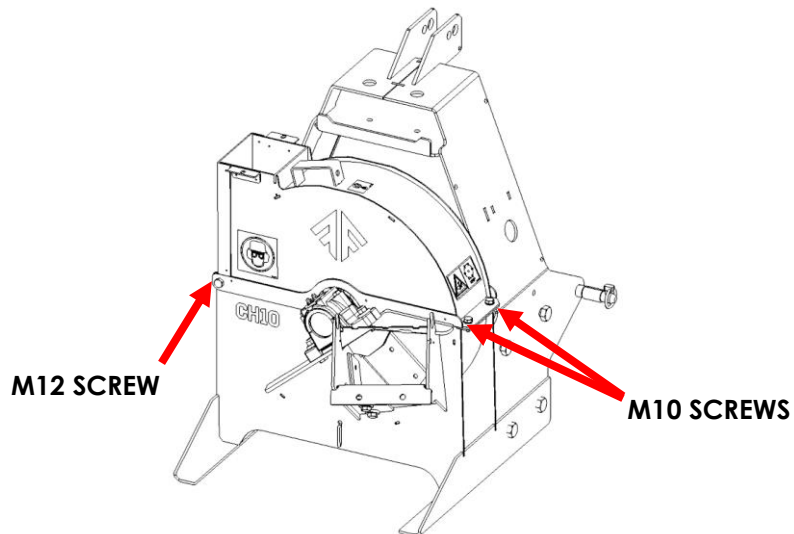
- Remove the knife fastening screws (M12). Turn the wrench in such a way that your hands would not hit the knife if the wrench should slip.



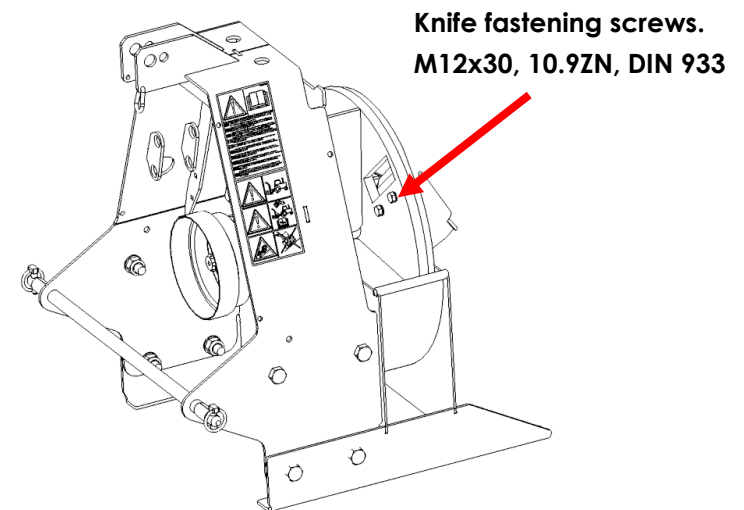
ATTENTION

- Wear protective gloves when handling knives or anvils.

UPPER FRAME FASTENING SCREWS



KNIFE FASTENING SCREWS



SHARPENING THE KNIVES



- Sharpen all knives equally. This ensures disc balance.
- Avoid heating the knife during sharpening.

The knives need sharpening when:

- The self-feeding of wood has decreased
- The power demand has increased
- The chip surface is rough
- The individual chips have become smaller.

Normally, the knives can be sharpened several times without actually being removed (with, e.g., a sharpening stone or belt grinder).

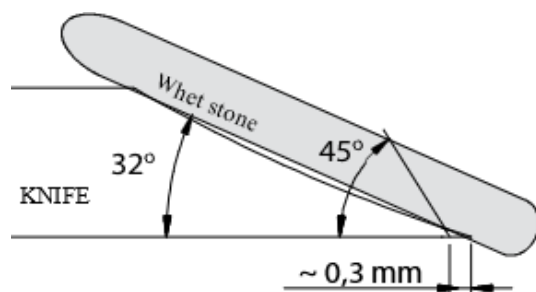
More thorough conditioning is carried out with a surface grinder, with the knives removed.

The grinding angle of the knives is 32° and their honing angle 45°. The honing angle is intended to prevent edge breakage.

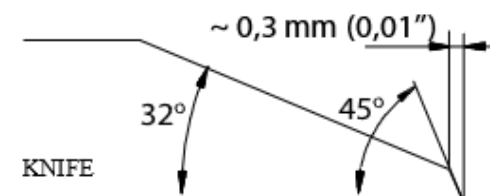
The hone angle is ground to a 45° angle with two to three longitudinal strokes, using a level sharpening stone.

Any burrs can be removed by parallel grinding of the surface opposite to the knife mounting screws.

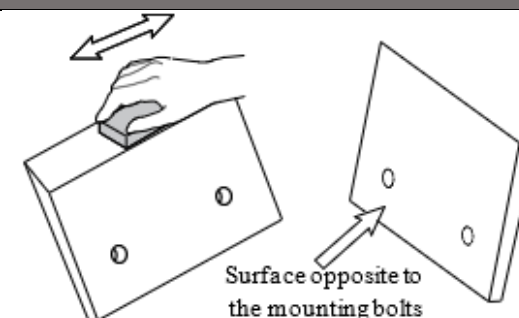
THE PROFILE OF A CONCAVE KNIFE



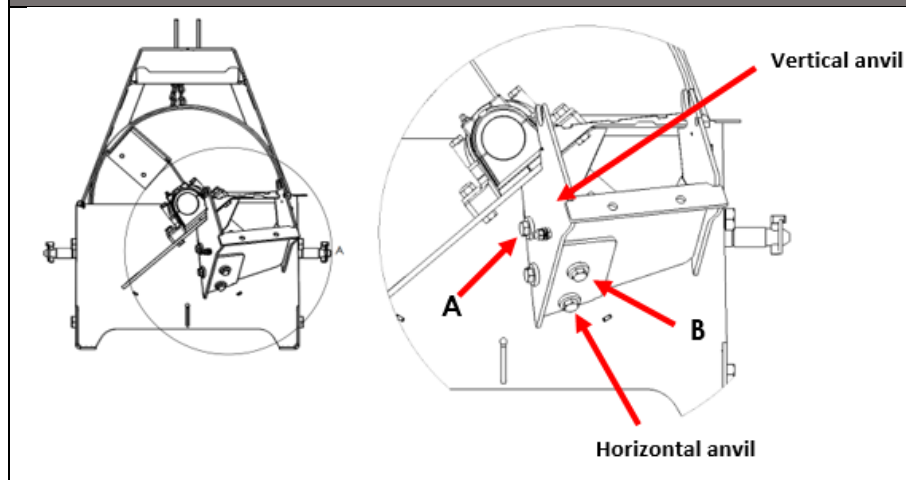
A KNIFE WITH A FLAT PROFILE



FINAL GRINDING OF THE KNIFE

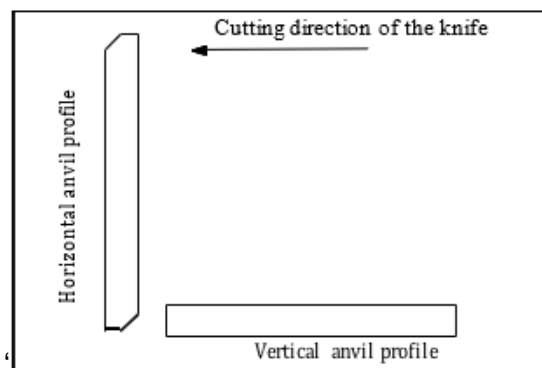


REMOVING THE ANVILS



The chipper features both a vertical and horizontal anvil. To remove the anvils, open the fastening screws (A) and (B) (M12). The horizontal anvil fastening screws (B) are located below the feed opening.

SHARPENING THE ANVILS

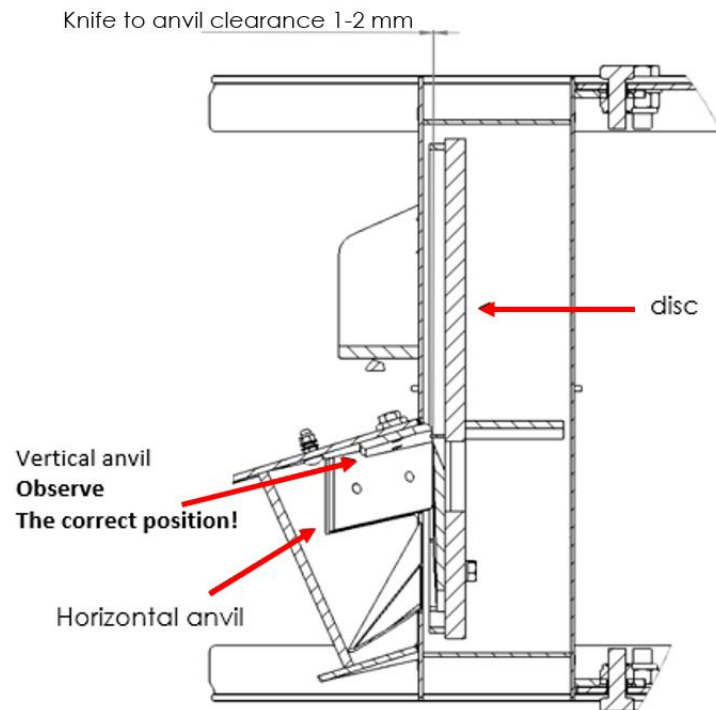


If you notice wear or rounding of the inner edge of the anvil, sharpen the anvils so that the original angles are retained.

INSTALLING THE KNIVES AND ANVILS

- Check the condition of the fastening screws and nuts.
- Install the knives and anvils and tighten the fastening screws to the torques specified in table.
- Adjust the knife-to-anvil clearance.

ADJUSTING AND CHECKING THE KNIFE-TO-ANVIL CLEARANCE



The need for adjusting the anvils is determined by the amount the knives are sharpened. Always check and, if necessary, adjust the clearance between knives and anvils

- after a heavy sharpening;
- if the knives were removed - for example, due to sharpening;
- if new knives are replaced,
- if chip length is adjusted.

Check the clearance with a feeler gauge.

1. Loosen the M12 locking screws (A) and (B) of the anvils. For markings see chapter: *Removing the anvils*.
2. Turn the cutting disc in such a way that the front edge of the knife is exactly opposite the vertical anvil. Slide a feeler gauge between the knives.
3. Tighten the screws of the anvil (A).
4. Adjust the play of the horizontal anvil towards the front edge of the knife to 1-2 mm.
5. Tighten the locking screw (B).
6. Recheck the knife play.

WARRANTY

The machine shall not be taken into use before the operating manual is thoroughly read and understood.

WARRANTY PERIOD:	Farmi Forest Oy (Farmi) is liable to defects which appear within a period of twelve (12) months.
NOTIFICATION OF DAMAGE:	- The purchaser shall without delay and in no case not later than two (2) weeks, notify Farmi or its authorized dealer about the defect. If the purchaser does not notify Farmi within two weeks, then the purchaser shall lose the right to have the defect remedied. - If there is a reason to believe that the defect may cause other damages, the operator shall cease using the product.
FARMI'S LIABILITY FOR DAMAGES:	- Farmi is liable only for defects which appear under the conditions of under proper use of the product. - Farmi's liability does not cover defects arising from faulty maintenance, incorrect operation and technical alternations carried out without Farmi's written consent. - Farmi's liability does not cover normal wear and tear or deterioration.
Farmi Forest Oy will not be responsible for any damages caused by the machine. Instructions and safety instructions in the manual must be read thoroughly and followed during every using session. The manufacturer will not assume any liability for any economical losses or for any consequential damage to persons or property, which may incur during product's usage or failure.	
Product warranty is NOT VALID in case of user's incompetence and/or irrespective handling of the equipment. Warranty does not cover transportation cost, repairing labor or loss of profit due to normal abrasion or product failure.	

Annex 1. Bolt tightening moments



- The information about bolt tightening moments using torque wrench is to be used as a guide only for standard metric threads that may not be suitable for all bolts.
- Described bolt tightening moments are meant only for standard sized ISO metric threads.
- Required tightening torques may vary depending on multiple factors like lubrication, surface condition etc.
- For more accurate tightening moments see VDI 2230 Standard.

Thread size	Strength class				
	4.6	5.8	8.8	10.9	12.9
M3	0.5				
M4	1.2				
M5	2.4	4,1	6.5		
M6	4.2	6.9	11		
M8	10	17	27		
M10	20	33	53	74	89
M12	34	57	91	128	154
M14	54	91	145	204	244
M16	83	139	222	313	375
M20	163	271	434	610	732
M22	219	365	584	822	986
M24	281	469	750	1050	1270

